



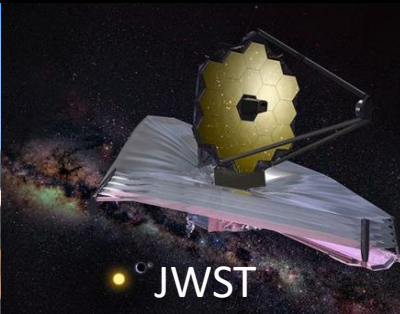
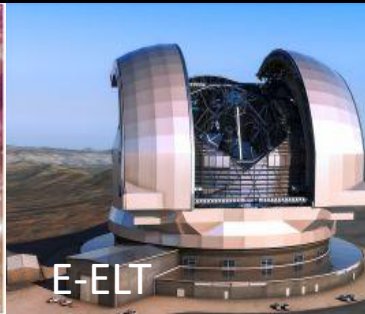
Kiara

HPC Challenges in next-generation
galaxy formation simulations

Romeel Davé, Doug Rennehan, Zhen Xiang,
and the Simba/Kiara team

Cosmological Simulations of Galaxy Formation

- *Goal:* Simulate a representative cosmic volume from the Big Bang until today, including all relevant physics, and have it yield an observationally concordant galaxy population across all of cosmic time.
- If we can do this, we can:
 - Have a realistic platform for running numerical experiments
 - Obtain a plausible set of physics that explains why galaxies look the way they do
 - Interpret observations spanning a wide range of wavelengths and cosmic epochs



Andromeda
Galaxy

X-ray

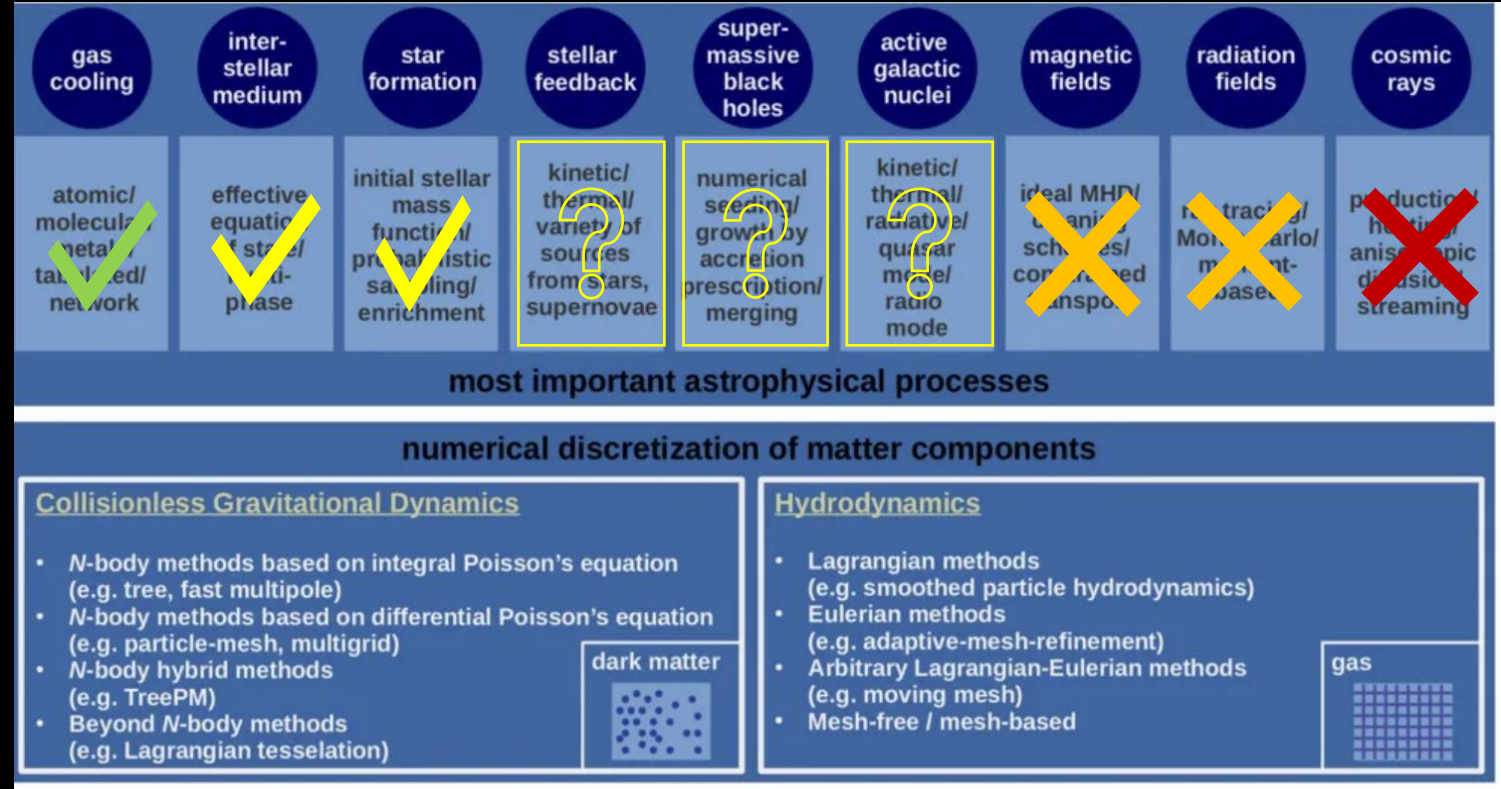
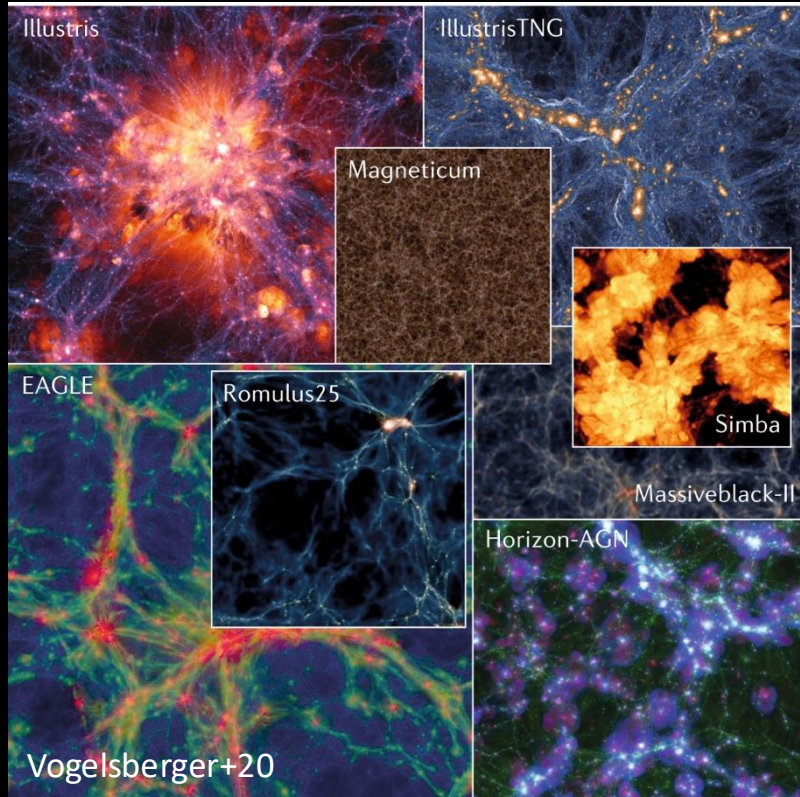
UV/Optical

Near/mid-IR

Far-IR/millimetre

Radio

Galaxy Formation Simulations: Input Physics



- *Typical size/resolution*: ~ 100 s Mpc box, ~ 1 kpc spatial resolution, $\sim 10^6 M_{\text{sun}}$ mass resolution, $\sim$ millions of galaxies
- Generally tuned to match *stellar* properties and $M_{\text{BH}}-M_*$.

IllustrisTNG

SIMBA

Ramses

CROCODILE



Temperature Slices

$z = 10.00$

CAMELS project: Temperature maps (*F. Villaescusa-Navarro*)

Magneticum

EAGLE

Astrid

Obsidian

The Simbaverse (so far)



- **Mufasa (2016):** Gizmo/MFM, Grackle-2, KMT H₂ SF, FIRE winds, AGN "feedback" via halo heating, no BH, no dust, 2×512^3 particles.
- **Simba (2019):** Grackle-3, +Torque/Bondi BH growth, +bipolar rad/jet/X-ray BH feedback, +dust evolution, 100 Mpc/h 2×1024^3 .
- **Spin-offs:**
 - **300 Gizmo-Simba runs (Cui+22):** Hundreds of cluster zooms
 - **CAMELS-Simba (V-N+23):** 1000's of 25 Mpc/h sims for machine learning
 - **Cosmic-Sands (Lower+23):** Massive EoR galaxy zooms
 - **Simba-C (Hough+2023):** Kobayashi chemistry, other improvements
 - **Simba-EoR (Jones+2024):** Improved ISM subgrid model with H₂, dust.
 - **Hyenas (Cui+24, Jennings+24):** Group zooms for IGrM, AGN feedback
- ***Next: KIARA!***



Kiara Physics

- Implemented in **SWIFT** for increased speed & scalability.
- **MAGMA-2 SPH** hydrodynamics
- New ISM subgrid model with **two-phase H_2 /dust/SF tracking**, in Grackle.
- Star formation based on **H_2 + lognormal density pdf** in dense ISM phase.
- Simba-C chemistry (**chem5 w/up to 34 elements**). Rennehan+21 diffusion.
- SF wind launching: **FIRE-like scalings**, as **SN energy comes available (w/ESF)**
- SF wind propagation: **Firehose model** for outflowing stream interactions.
- BH accretion: Torque-limited for cold gas, Bondi for $T > 10^5$ gas, **variable f_{acc}** .
- BH feedback: **Obsidian 3-mode** (ADAF, quasar, slim disk) based on f_{Edd} .

*New features compared with Simba

MAGMA-2 SPH

Rosswog (2020), implementation in SWIFT by D. Rennehan & Z. Xiang incl. cosmology

- SPH: Navier-Stokes solver in adaptive Lagrangian frame, no grid.
- New in Magma-2:

- Artificial viscosity uses midpoint quantities and slope limiters to reduce unwanted dissipation:

$$\Pi_{ib} = \begin{cases} \frac{-\alpha \bar{c}_{ib} \mu_{ib}}{\bar{\rho}_{ib}} & \text{if } \vec{v}_{ib} \cdot \vec{r}_{ib} < 0 \\ 0 & \text{otherwise} \end{cases}$$

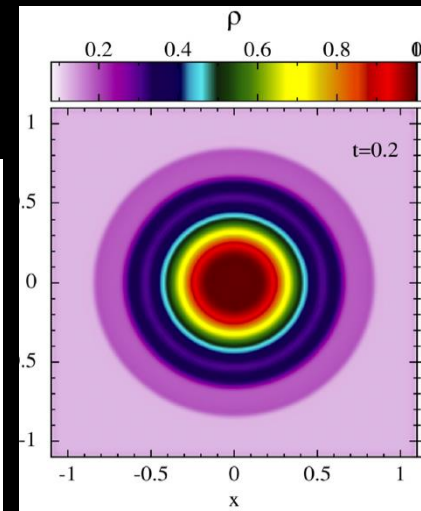
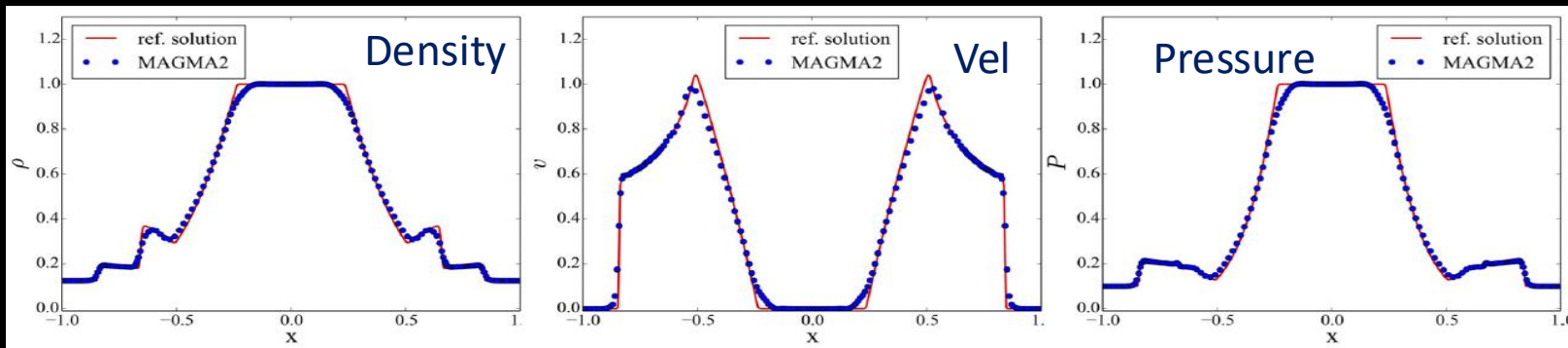
Where $\mu_{ib} = \frac{h \vec{v}_{ib} \cdot \vec{r}_{ib}}{r_{ib}^2 + \eta^2}$, with $\vec{v}_{ib}$ representing the *reconstructed* velocity differences,

- Kernel gradients computed via a correction matrix representing spatial distribution of neighbors:

$$\nabla A_i = \mathbf{C}_i \cdot \sum_b m_b (A_b - A_i) \nabla W_{ib}$$

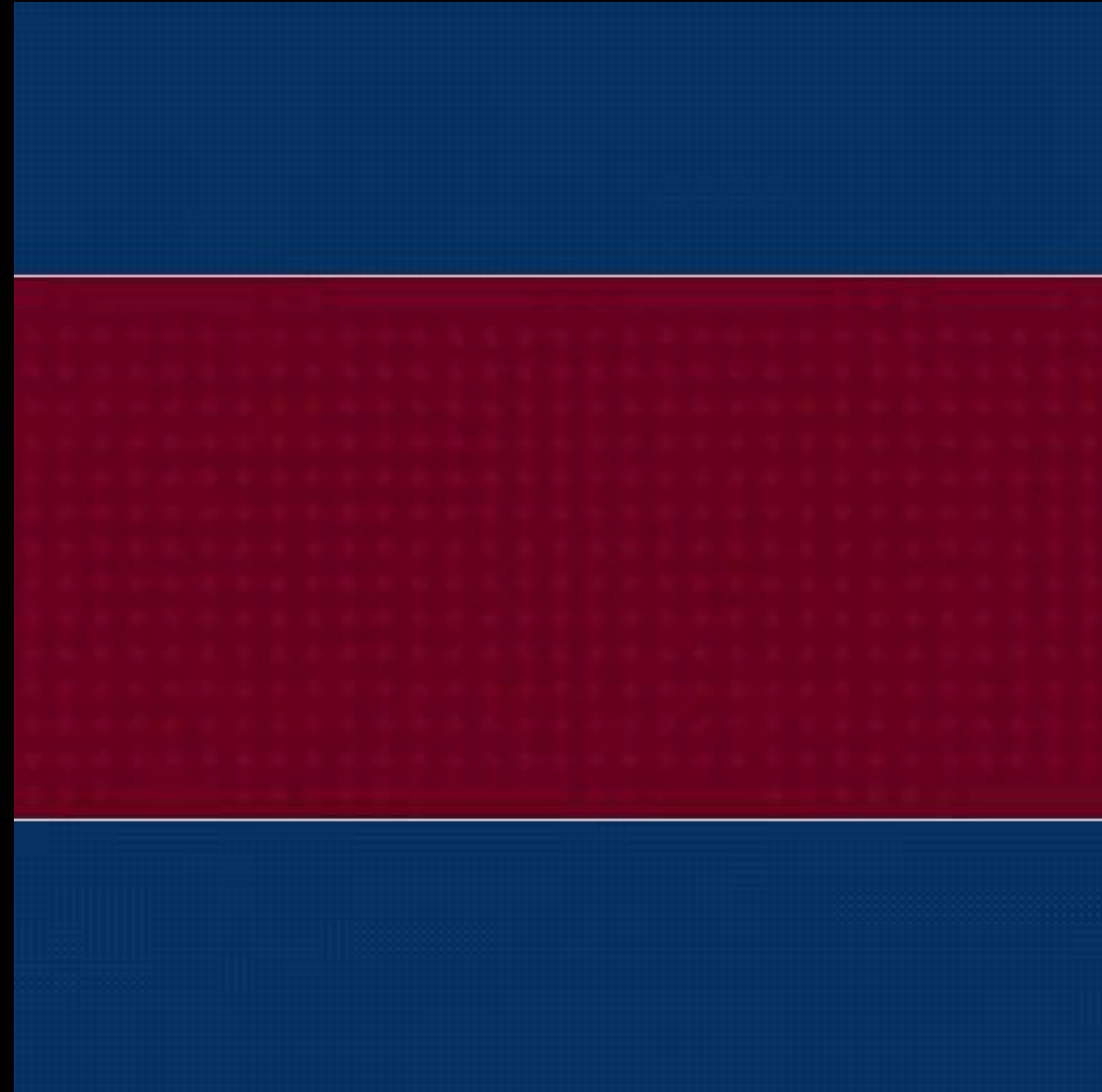
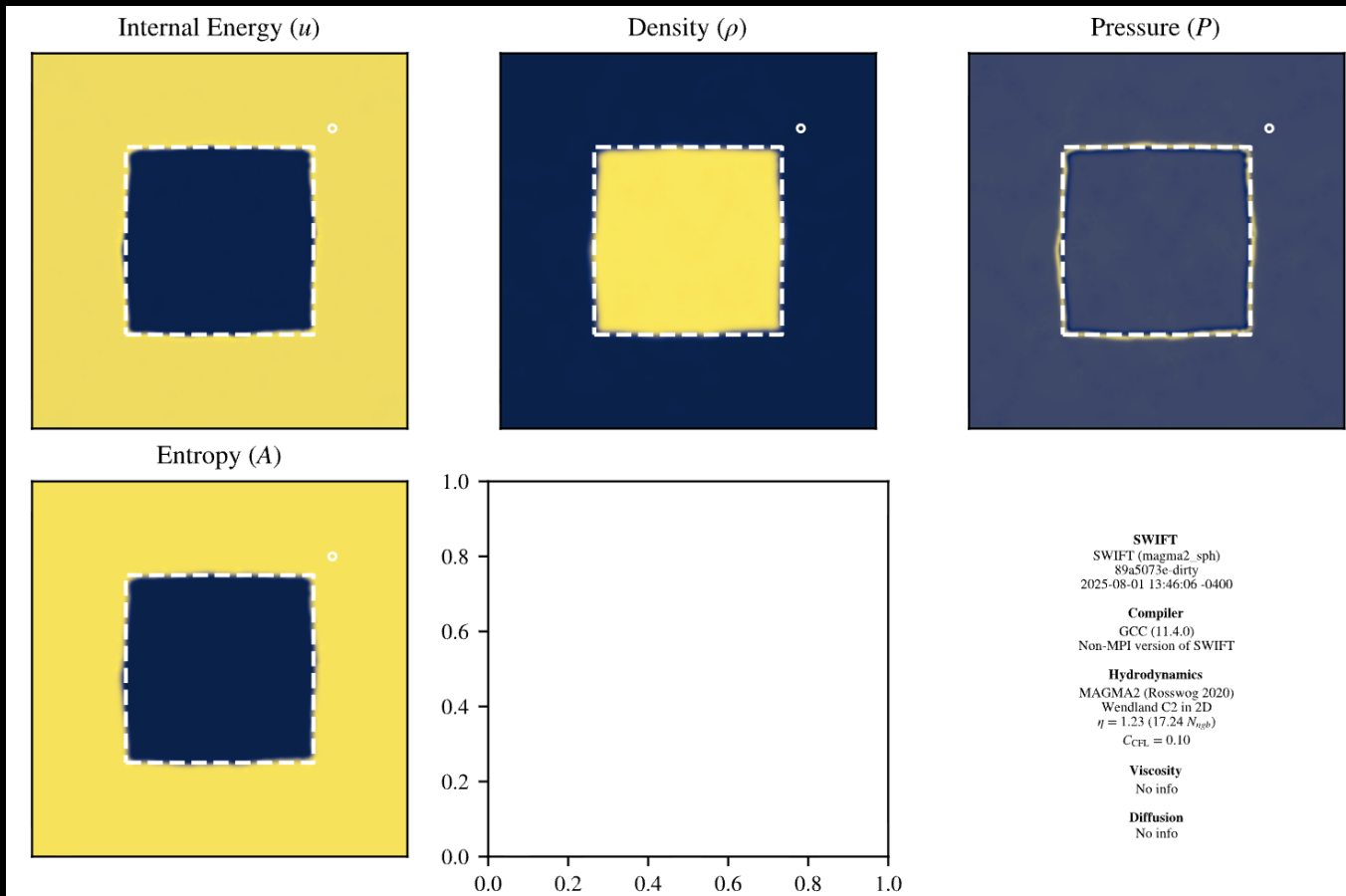
- Slower! Needs 114 neighbors, matrix inversion to get $\mathbf{C}_i$.

Spherical blast wave
with Magma-2



MAGMA-2 SPH

implementation by D. Rennehan



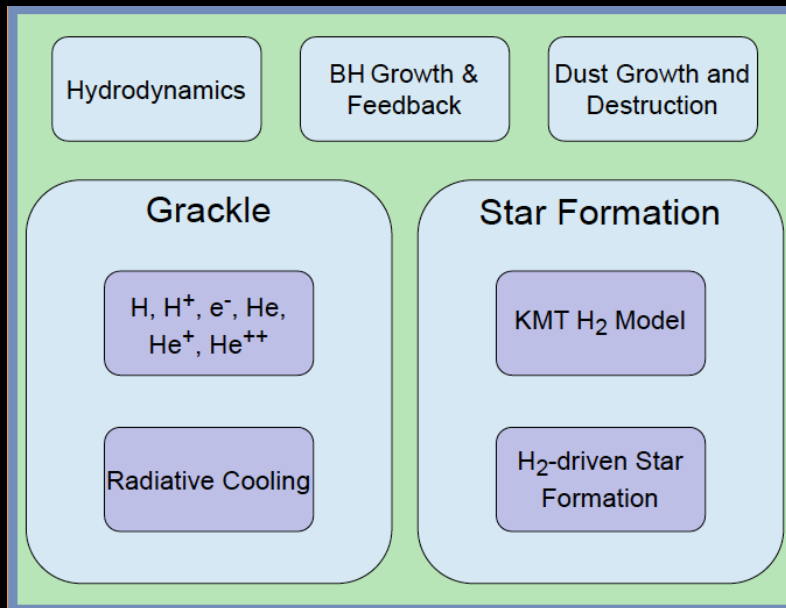
Note: The exact same kernel & viscosity setting were used for these tests as in Kiara

Simba-EoR's subgrid ISM model (Jones,Smith,RD 24)

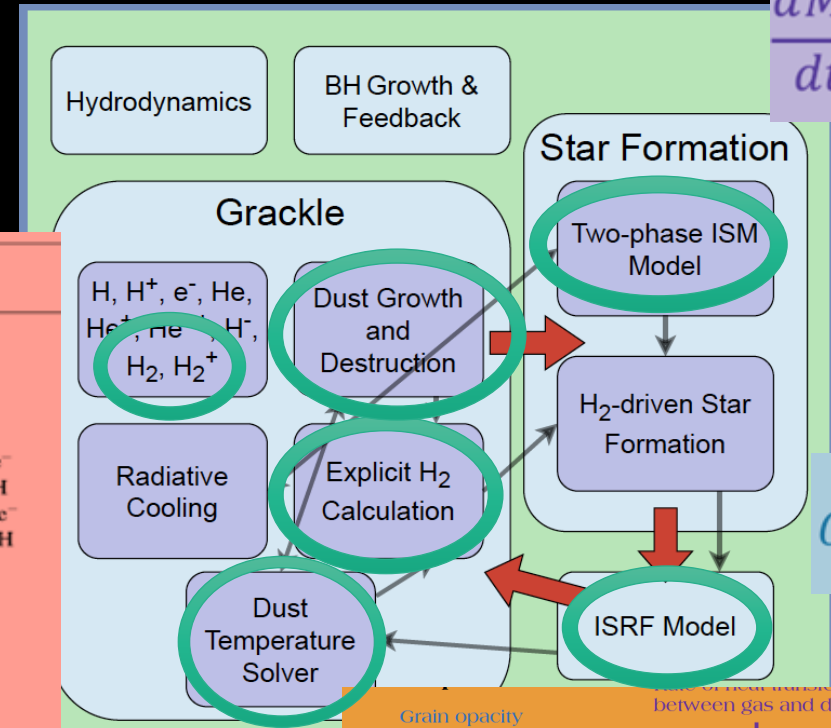
- Self-consistently evolve atomic gas, molecular gas, dust, and star formation in Grackle, with ISRF parameterized from FIRE/Latte sims.
- ISM gas: Warm 10^4K phase in pressure equil w/evolved cold phase ($T < \sim 100$).

SIMBA: Pressurized 1-phase

KIARA: Two-phase



Reaction		
$\text{H} + \text{e}^-$	$\rightarrow$	$\text{H}^- + \gamma$
$\text{H}^- + \text{H}$	$\rightarrow$	$\text{H}_2 + \text{e}^-$
$\text{H} + \text{H}^+$	$\rightarrow$	$\text{H}_2^+ + \gamma$
$\text{H}_2^+ + \text{H}$	$\rightarrow$	$\text{H}_2 + \text{H}^+$
$\text{H}_2 + \text{H}^+$	$\rightarrow$	$\text{H}_2^+ + \text{H}$
$\text{H}_2 + \text{e}^-$	$\rightarrow$	$\text{H} + \text{H} + \text{e}^-$
$\text{H}_2 + \text{H}$	$\rightarrow$	$\text{H} + \text{H} + \text{H}$
$\text{H}^- + \text{e}^-$	$\rightarrow$	$\text{H} + \text{e}^- + \text{e}^-$
$\text{H}^- + \text{H}$	$\rightarrow$	$\text{H} + \text{e}^- + \text{H}$
$\text{H}^- + \text{H}^+$	$\rightarrow$	$\text{H} + \text{H}$
$\text{H}^- + \text{H}^+$	$\rightarrow$	$\text{H}_2^+ + \text{e}^-$
$\text{H}_2^+ + \text{e}^-$	$\rightarrow$	$\text{H} + \text{H}$
$\text{H}_2 + \text{H}^-$	$\rightarrow$	$\text{H}_2 + \text{H}$
$\text{H} + \text{H} + \text{H}$	$\rightarrow$	$\text{H}_2 + \text{H}$
$\text{H} + \text{H} + \text{H}_2$	$\rightarrow$	$\text{H}_2 + \text{H}_2$
$\text{H}^- + \gamma$	$\rightarrow$	$\text{H} + \text{e}^-$
$\text{H}_2^+ + \gamma$	$\rightarrow$	$\text{H} + \text{H}^+$
$\text{H}_2 + \gamma$	$\rightarrow$	$\text{H}_2^+ + \text{e}^-$
$\text{H}_2^+ + \gamma$	$\rightarrow$	$\text{H}^+ + \text{H}^+ + \text{e}^-$
$\text{H}_2 + \gamma$	$\rightarrow$	$\text{H} + \text{H}$
$\text{H} + \text{H} + \text{grain}$	$\rightarrow$	$\text{H}_2 + \text{grain}$



$$\frac{dM_*}{dt} = \epsilon_* \frac{\rho f_{\text{H}_2}}{t_{\text{dyn}}}$$

$$G_0 = \frac{G_{0,MW}}{\Sigma_{MW}} \Sigma$$

$$4\sigma T_{gr}^4 \kappa_{gr} = 4\sigma T_{CMB}^4 \kappa_{gr} + \Lambda_{gas-grain}$$

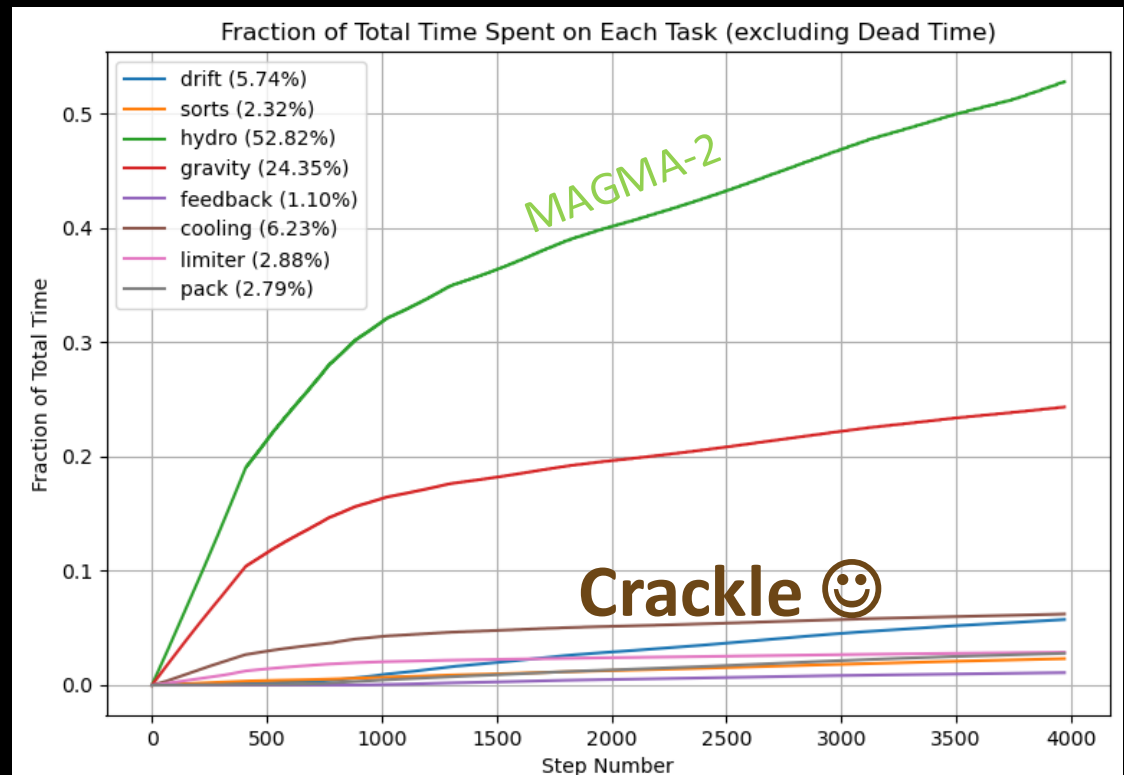
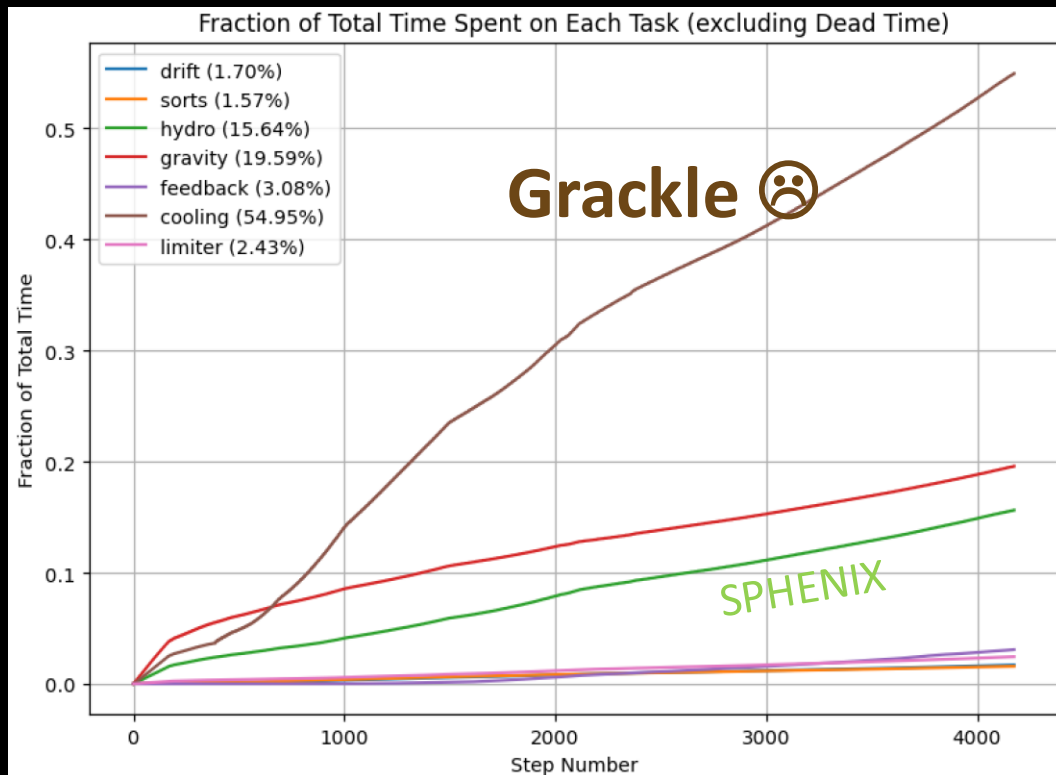
Radiative cooling Heating

Jones, Smith, RD 2024

Crackle: C version of Grackle*

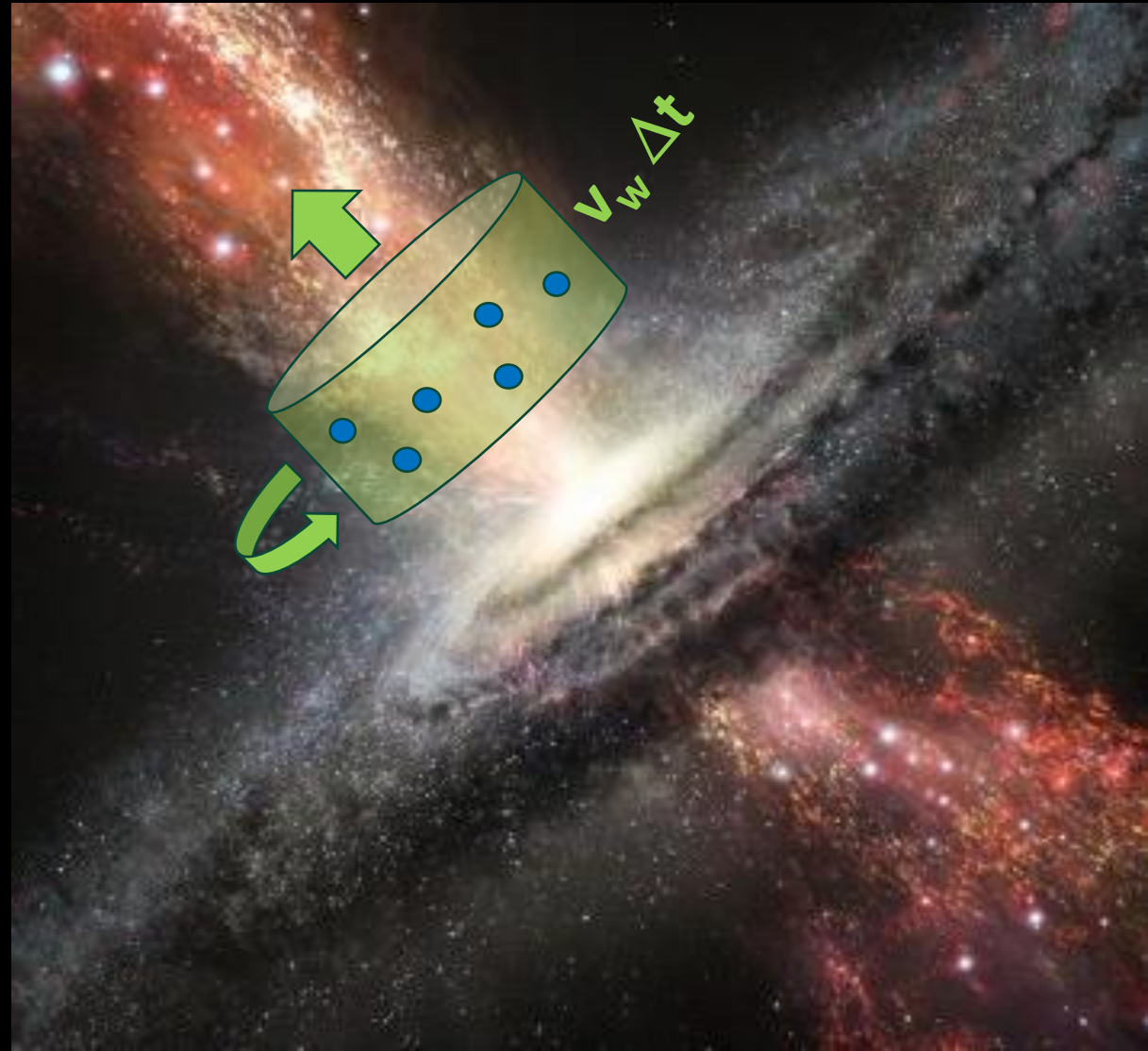
*only with the options needed for Kiara, and only works with the Grackle kiara branch.

- Crackle (<https://github.com/romeeld/crackle>) provides drop-in replacements and add-ons, no change to API just set `use_grackle=2`.
- Same answer for a single substep, but different substepping scheme.
- Primary advantage is speed, also debuggability.



SF Feedback: Kinetic ejection, Firehose wind propagation

- SN winds ejected only once enough SN have gone off.
- Eject kinetic winds, with mass loading & wind speed roughly based on FIRE, tuned.
- **Firehose model:** No decoupling! Outflows treated as coherent stream of gas moving through ISM/CGM, using formalism from Mandelker+20.
- Exchange momentum, metals with ambient gas; excess E heats gas.



Kiara is born!

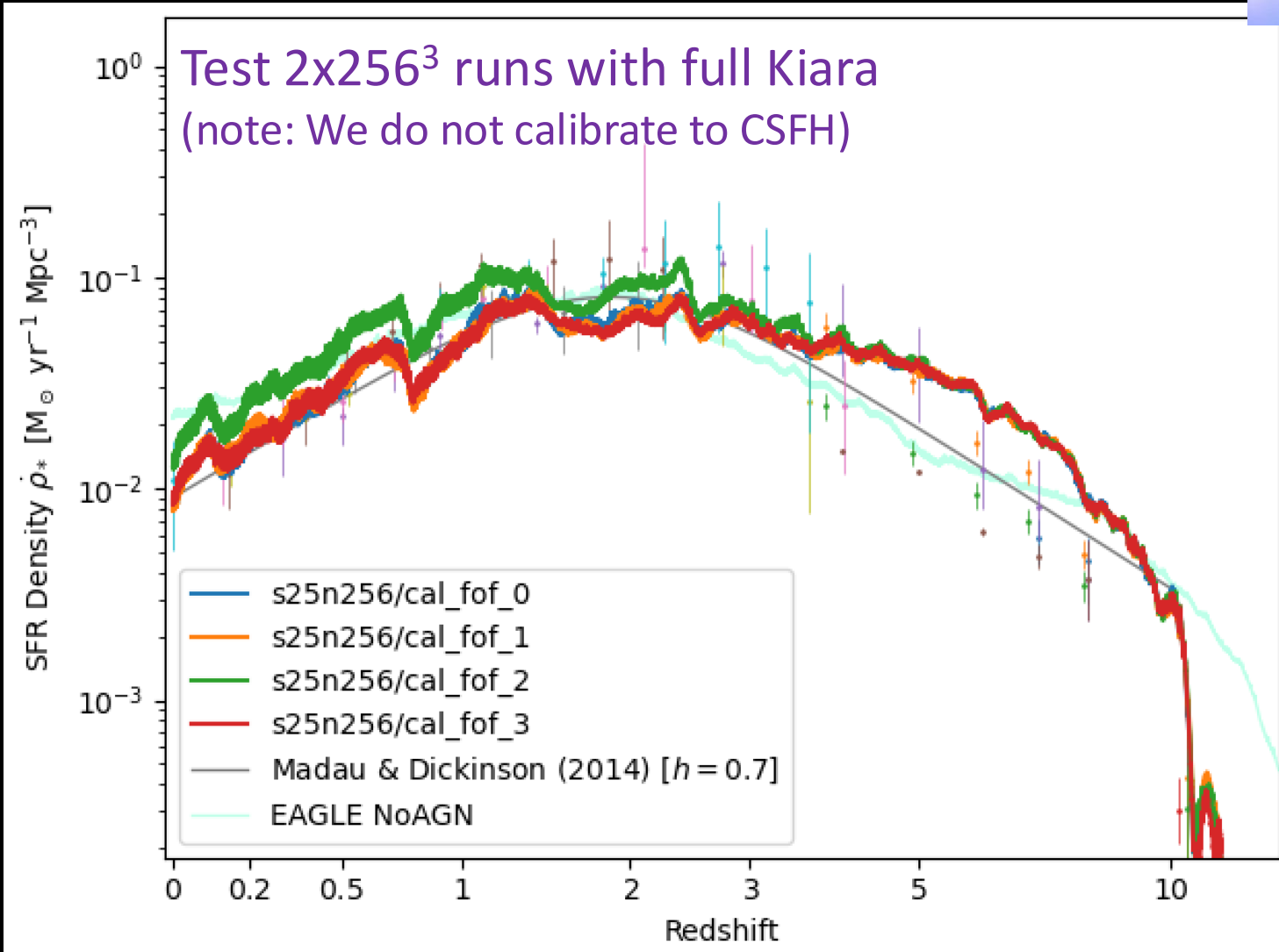


Runs efficiently, $>\sim 2\text{-}3\times$ faster than GIZMO, with a lot more physics.

Does not crash (mostly)

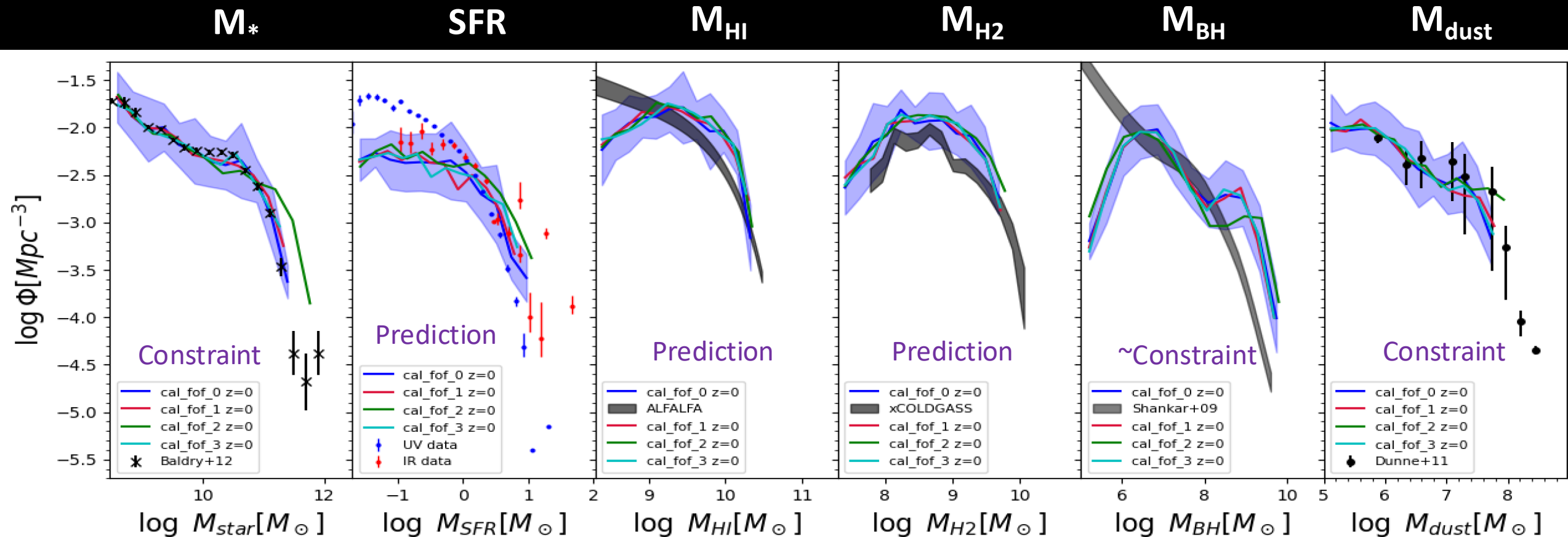
Reasonable galaxies & SFRD evolution.

Some subgrid models still need tweaking



Mass functions in various quantities vs data

KIARA 25 Mpc/h, 2×256^3 particles @z=0



Analysis: Caesar → Octavian

Jakub Szpila, JP Duminy, Kevin Lo, Romeel Davé

- Caesar (caesar.readthedocs.io) inputs a halo catalog & runs a 6DFOF to find galaxies, pre-computes 100's of properties incl. photometry. Outputs easy-to-use cross-matched hdf5 catalogs for each snapshot.
- Octavian: Ground-up rewrite of Caesar. Speed up by over an order of mag, peak memory usage down by x5, MPI-scalable. Same output.
- Octavian adds support for various halo finders (AHF, HBT), subhalos, modular processing, importing of merger trees.
- Alpha version working, beta release expected by end of summer.
- SWIFT (swiftsimio) support coming, integrated with swift-galaxy



Give me 5 hrs
and I shall
compute the
Universe!



Bruh. I did it
in like, 20
minutes.



Summary

- Kiara is the SWIFT-based successor to Simba(-C), with lot of improved and redesigned input physics:
 - New subgrid ISM/dust model + SF model + chem5
 - New wind interaction model (no “decoupling”)
 - Lots of other novel physics modules
- Still to be finished: Dust grain size distribution, BH spin tracking, ML-based modeling of ISRF.
- Reasonable-looking galaxies, but some calibration still needed. Massive galaxies currently not quenching early enough.
- Preliminary production runs hopefully this Fall! (fingers crossed)